

Pre-Requisite(s): 23CSEXXX Computer Networks

Course Objectives

- This course helps students to acquire a solid understanding of cloud computing fundamentals, including service and deployment models, to make informed decisions when selecting and implementing cloud-based solutions for various projects and scenarios.
- Learn and apply modern cloud-native application development techniques, such as microservices, containerization, and container orchestration, to build scalable, resilient, and efficient applications that leverage the full potential of cloud environments.
- Develop practical skills in managing and monitoring cloud deployments, focusing on orchestration, automation, and resource management, enabling students to optimize and maintain cloud-based solutions for improved performance, cost-effectiveness, and overall operational efficiency.

Course Outcomes

CO1: Understanding of core cloud computing concepts, service models, and deployment models, along with the ability to identify the benefits and challenges associated with cloud computing.

CO2: Proficiency in utilizing cloud infrastructure and services with the ability to design and manage cloud resources effectively.

CO3: Ability to develop and deploy cloud applications by incorporating knowledge of compute, storage, and networking services.

CO4: Competence in cloud-native application development, including the design and implementation of microservices, containerization with Docker, and container orchestration.

CO5: Proficiency in managing and monitoring cloud environments, encompassing cloud orchestration, automation, and resource management techniques.

CO-PO Mapping

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO														
CO1	3		2	1	3								2	2
CO2	3	3	3	2	3				2				3	3
CO3	3	3	3	2	3				2				3	3
CO4	3	3	3	2	2				2				3	3
CO5	3	3	2	2	3				2				3	3

Syllabus

Unit-1

Introduction to Cloud Computing - Definition and evolution of cloud computing - Cloud computing service models: IaaS, PaaS, SaaS - Cloud deployment models: Public, Private, Hybrid, and Community Clouds - Benefits and challenges of cloud computing. Cloud Infrastructure and Services - Cloud computing architecture - Virtualization concepts - Storage solutions - Cloud networking and security.

Unit-2

Infrastructure as a Service (IaaS) - IaaS concepts and components - Compute services - Storage services - Networking services - Security and identity services. Platform as a Service (PaaS) - PaaS concepts and components - Application development platforms - Data management services - Middleware services - Integration and deployment services. Software as a Service (SaaS) - SaaS concepts and components - SaaS application examples - SaaS development and delivery.

Unit-3

Cloud Management and Monitoring - Cloud orchestration and automation - Cloud resource management. Cloud-Native Application Development - Microservices architecture - Monolithic vs. microservices architecture - Designing and implementing microservices. Containerization - Docker: container images, registries, and runtimes - Containerizing applications and managing dependencies - Multi-container applications and Docker Compose. Container Orchestration – Overview of platforms - Kubernetes: architecture, components, and concepts - Deploying and managing applications in Kubernetes - Service discovery, scaling, and rolling updates - Alternative orchestration platforms: Docker Swarm.

Textbook(s)

Thomas Erl, Zaigham Mahmood, and Ricardo Puttini, “Cloud Computing: Concepts, Technology & Architecture”, Prentice Hall, 2013.

Tom Laszewski, Kamal Arora, Piyum Zonooz, and Erik Farr, “Cloud Native Architectures: Design high-availability and costeffective applications for the cloud”, Packt Publishing, 2018.

Reference(s)

Toby Velte, Anthony Velte, and Robert Elsenpeter, “Cloud Computing: A Practical Approach”, McGraw-Hill Education, 2009.

Michael J. Kavis, “Architecting the Cloud: Design Decisions for Cloud Computing Service Models (SaaS, PaaS, and IaaS)”, Wiley, 2014.

Evaluation Pattern: 70:30

Assessment	Internal	End Semester
Midterm	20	
Continuous Assessment Theory (*CAT)	10	
Continuous Assessment Lab (*CAL)	40	
**End Semester		30 (50 Marks; 2 hours exam)

*CAT – Can be Quizzes, Assignments, and Reports

*CAL – Can be Lab Assessments, Project, and Report

**End Semester can be theory examination/ lab-based examination/ project presentation

GENERAL INFORMATION ABBREVIATIONS

USED IN THE CURRICULUM

L - Lecture

T - Tutorial

P - Practical

C - Credits

ENGG - Engineering Sciences (including General, Core and Electives)

HUM - Humanities (including Languages and others)

PRJ - Project Work (including Seminars)

CSE - Computer Science and Engineering

Course Outcome (CO) – Statements that describe what students are expected to know, and are able to do at the end of each course. These relate to the skills, knowledge and behaviour that students acquire in their progress through the course.

Program Outcomes (POs) – Program Outcomes are statements that describe what students are expected to know and be able to do upon graduating from the Program. These relate to the skills, knowledge, attitude and behaviour that students acquire through the program. NBA has defined the Program Outcomes for each discipline.

PROGRAM OUTCOMES FOR ENGINEERING:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO):

- Graduate will strive on a global platform to pursue their professional career in Computer Science and Engineering.
- Graduate will contribute to product development as entrepreneurs in inter-disciplinary fields of engineering and technology.
- Graduate will demonstrate high regard for professionalism, integrity and respect values in diverse culture, and have a concern for society and environment.

PROGRAM SPECIFIC OUTCOMES (PSO):

- Ability to design and implement innovative, optimal and elegant computing solutions to interdisciplinary problems using standard practices, tools and technologies.
- Ability to learn emerging computing paradigms for research and innovation